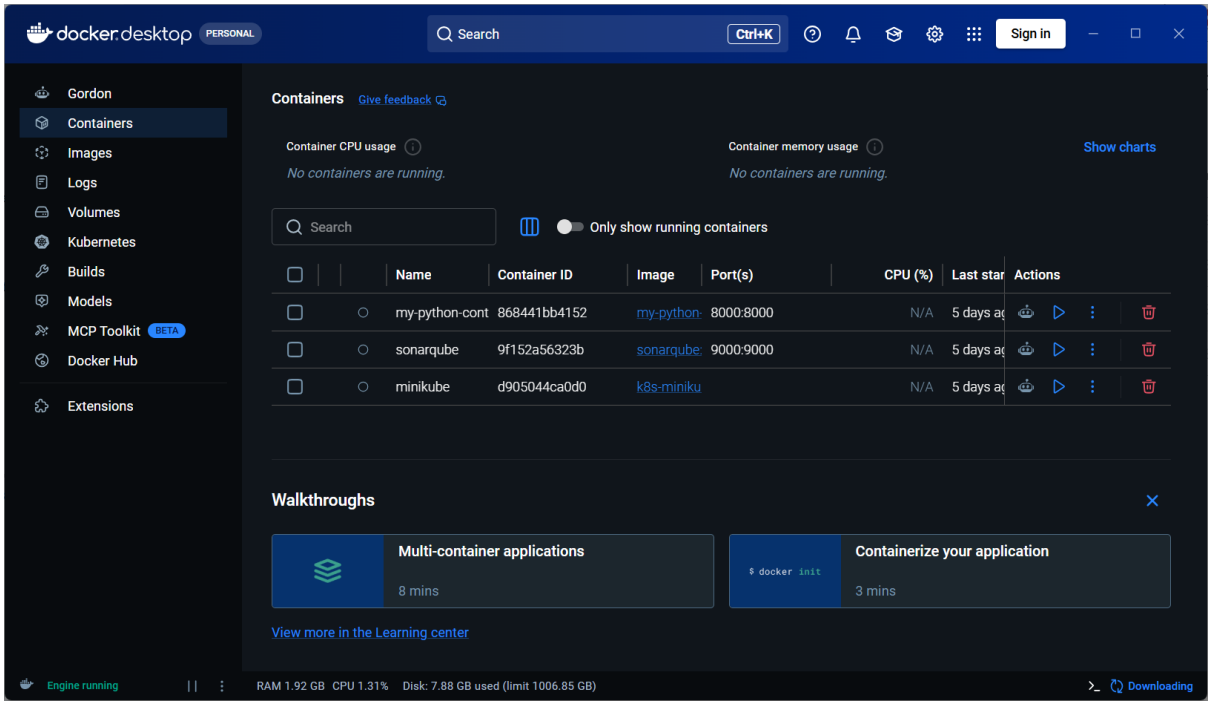
Week-6 ASE

Step 1: Check Docker Desktop

First, **open Docker Desktop** on your computer.

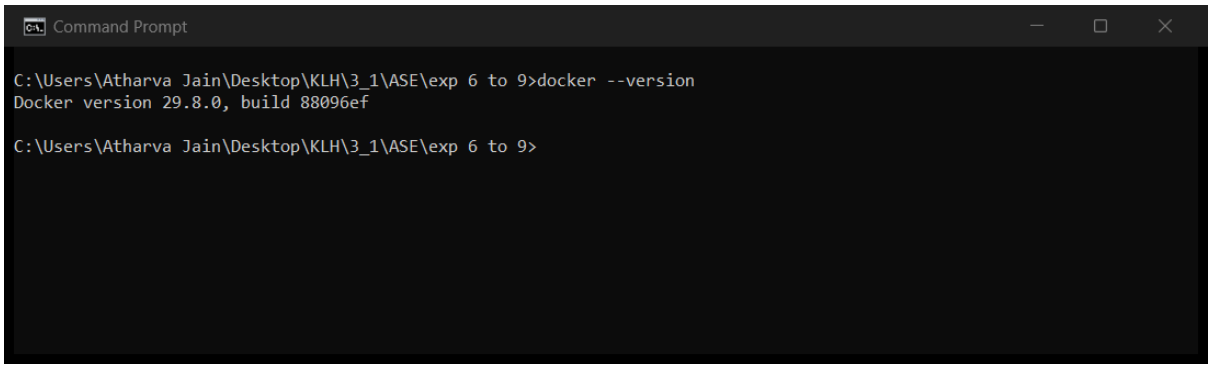
Wait until Docker Desktop shows that it is running.



Then open **PowerShell**.

Run:

docker --version



Step 2: Create a project folder

In PowerShell, go to a location where you want to create the project.

For example:

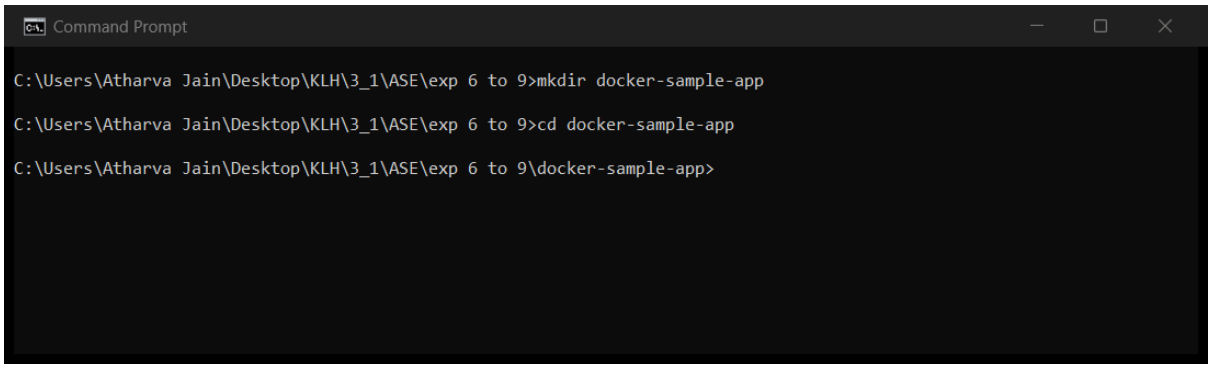
cd "C:\Users\Atharva Jain\Desktop\KLH\3_1\ASE\exp 6 to 9"

Create a folder:

mkdir docker-sample-app

Go inside it:

cd docker-sample-app



Step 3: Create index.html

Make sure your prompt looks like:

C:\Users\Atharva Jain\Desktop\KLH\3_1\ASE\exp 6 to 9\docker-sample-app>

Now type:

notepad index.html

Press **Enter**.

Notepad will open and may ask:

Do you want to create a new file?

Click **Yes**.

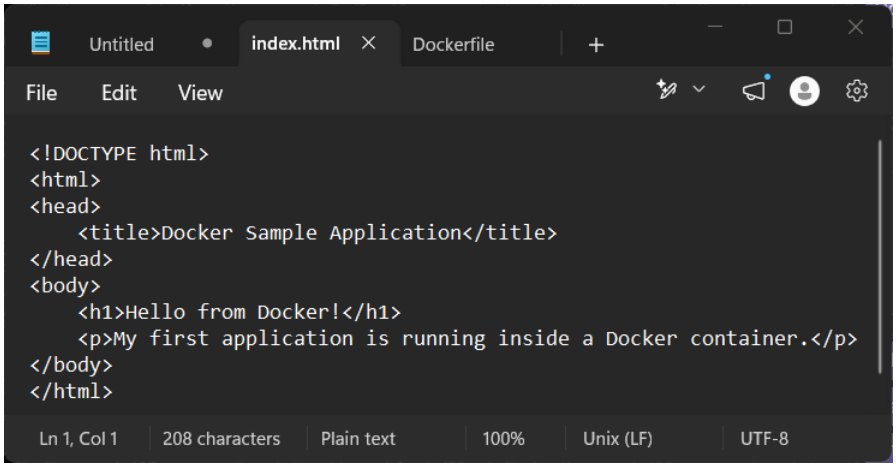
Paste this exactly:

```
<!DOCTYPE html>
<html>
<head>
  <title>Docker Sample Application</title>
</head>
<body>
  <h1>Hello from Docker!</h1>
  <p>My first application is running inside a Docker container.</p>
</body>
</html>
```

Then:

File → Save

Close Notepad.



Step 4: Check that the file was created

Back in CMD, type:

dir

You should see:

index.html

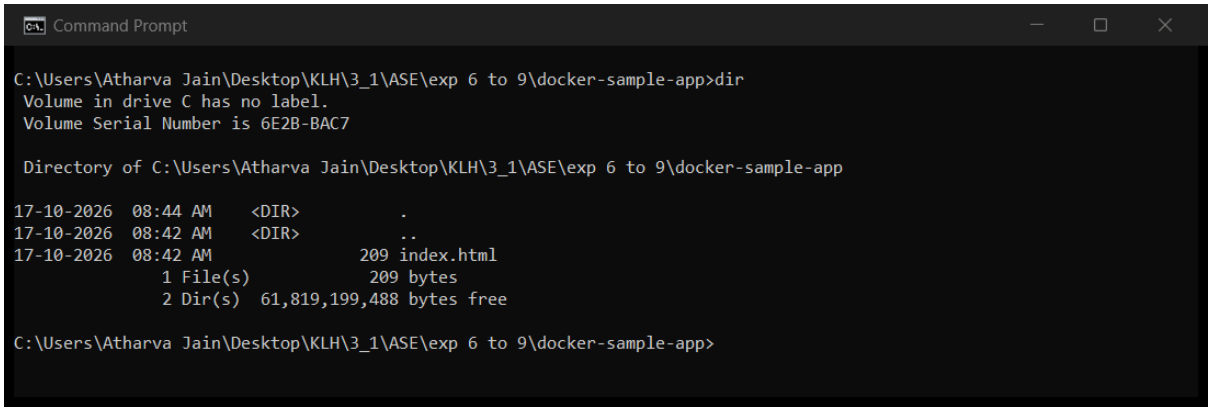
⚠️ **Important**

Make sure the file is named:

index.html

not

index.html.txt



Step 5: Create the Dockerfile

Type:

notepad Dockerfile

Press **Enter**.

If Notepad asks whether you want to create a new file, click **Yes**.

Step 6: Paste this Dockerfile

Copy and paste exactly:

FROM nginx:latest

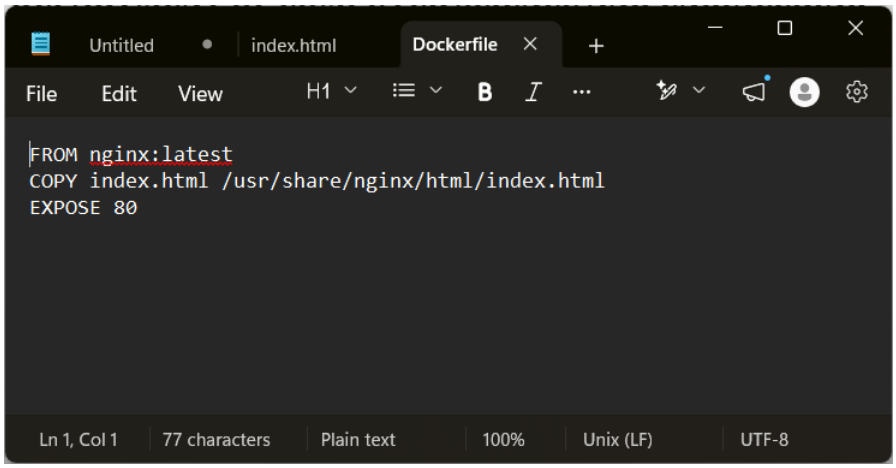
COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

Then:

File → Save

Close Notepad.



Step 7: Check the files

Back in CMD, run:

dir

You should see **both**:

Dockerfile

index.html

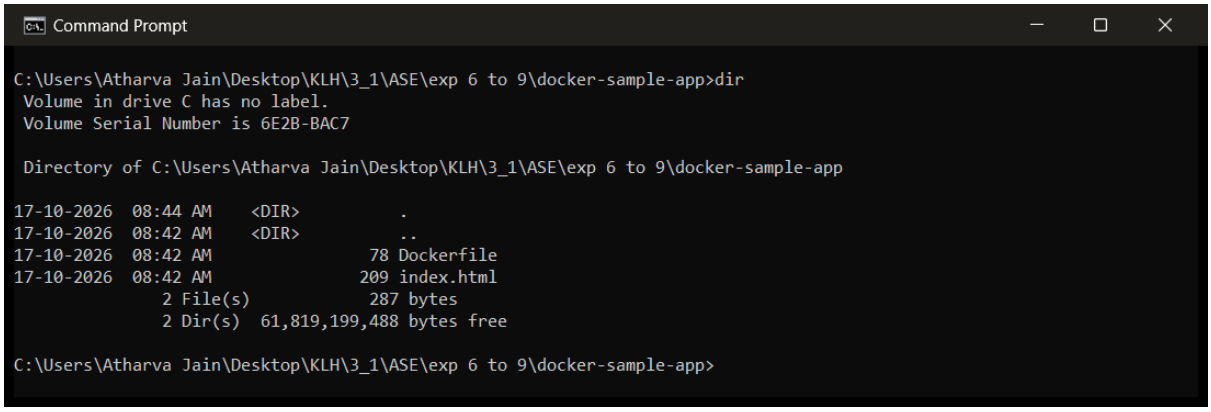
Your folder should now contain:

docker-sample-app

│

├── Dockerfile

└── index.html



Step 8: Make sure you are in the project folder

Your CMD should show:

C:\Users\Atharva Jain\Desktop\KLH\3_1\ASE\exp 6 to 9\docker-sample-app>

If yes, run:

docker build -t docker-sample-app .

⚠️ **Important:** Keep the . at the end.

Step 9: Wait for the build to finish

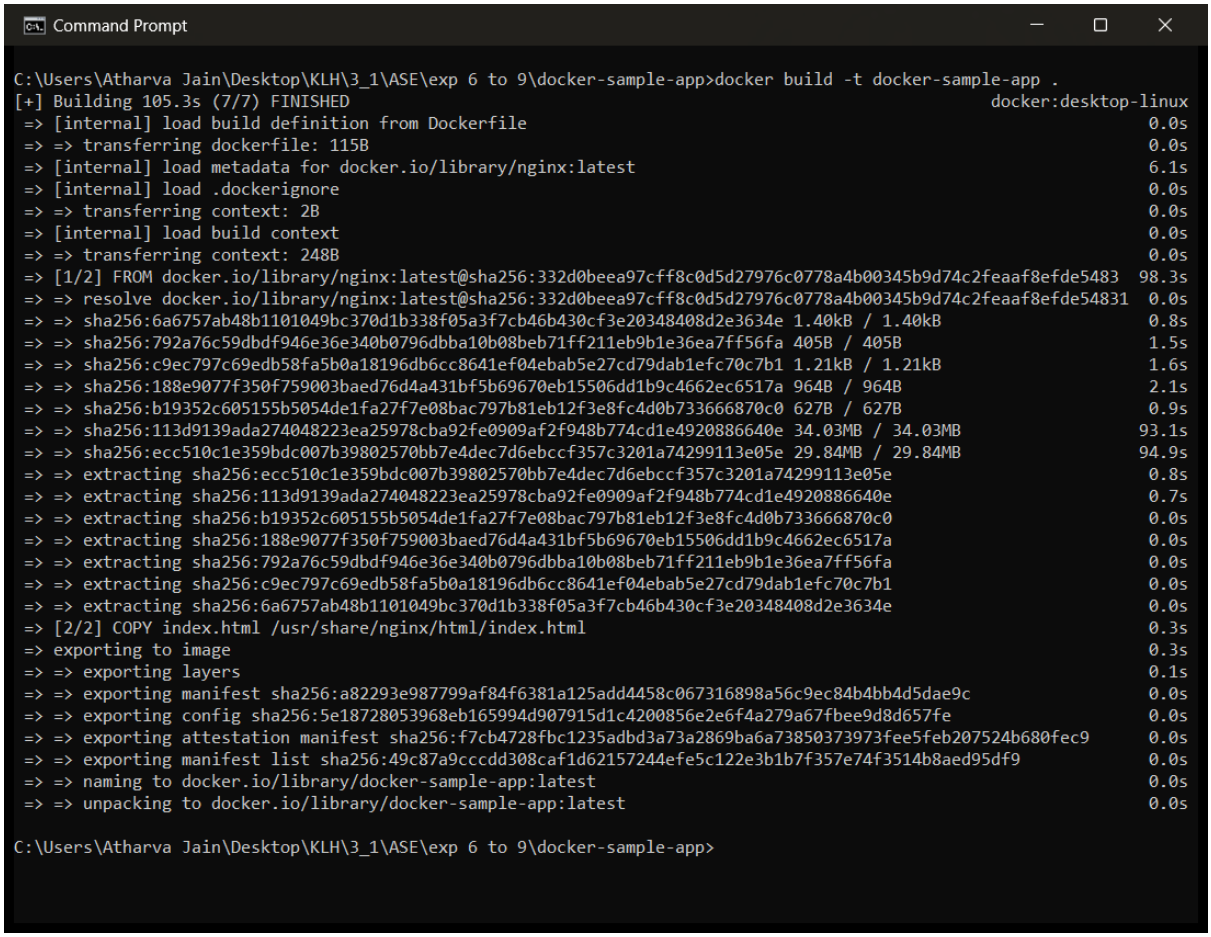
You should see Docker doing things like:

[+] Building ...

[+] ...

At the end, you should see something similar to:

Successfully tagged docker-sample-app:latest

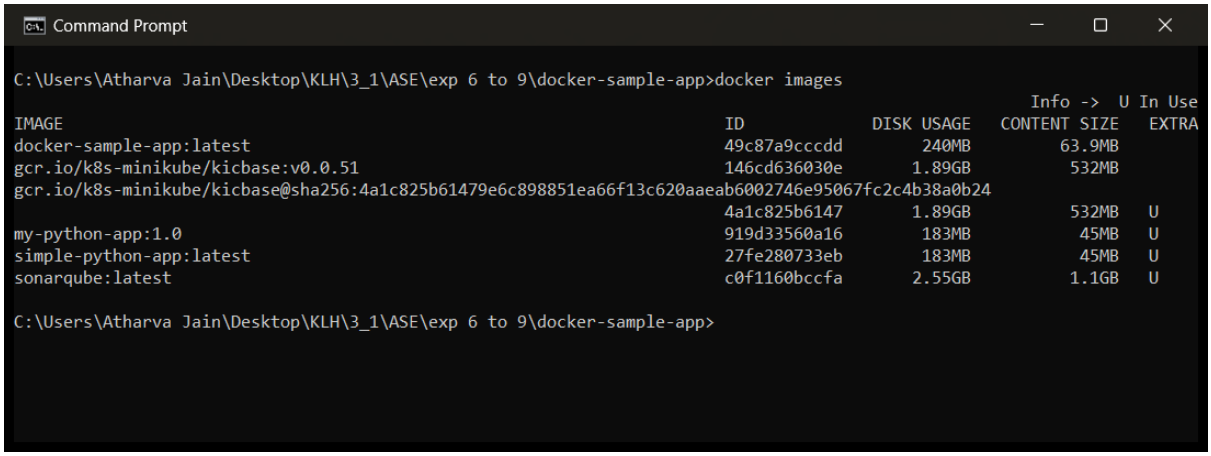


Step 10:Then run:

docker images

You should see:

docker-sample-app latest



Step 11: Run the container

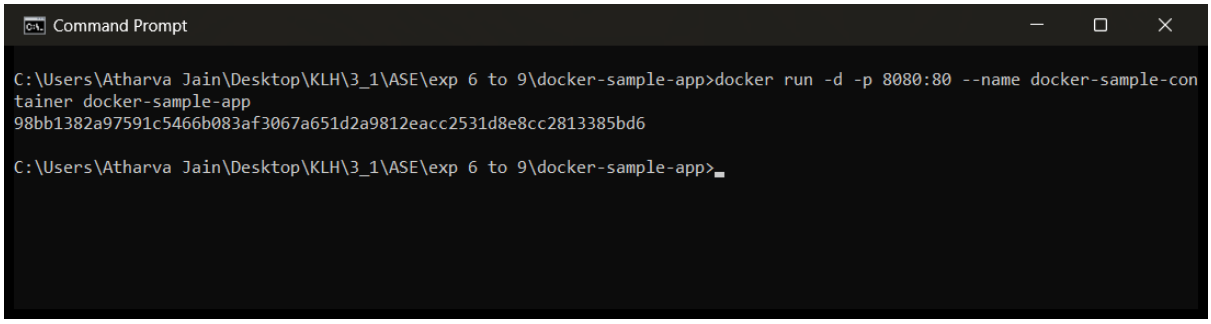
Run this **exactly**:

docker run -d -p 8080:80 --name docker-sample-container docker-sample-app

If successful, Docker will display a long container ID, for example:

a7c8d9e0f1...

That means the container has started.



Step 12: Check that the container is running

Run:

docker ps

You should see something similar to:

CONTAINER ID IMAGE STATUS PORTS

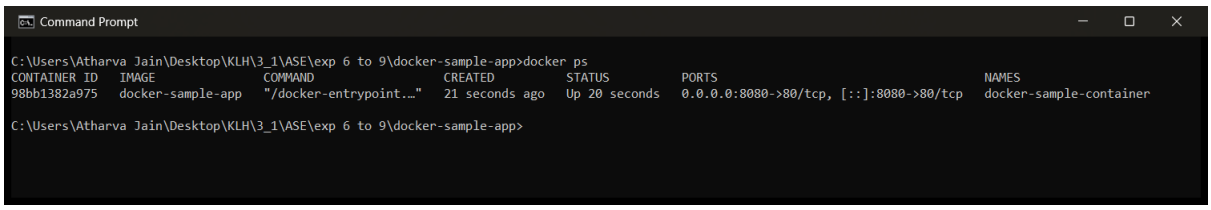
xxxxxxxx docker-sample-app Up ... 0.0.0.0:8080->80/tcp

The important things are:

IMAGE: docker-sample-app

STATUS: Up

PORTS: 8080->80



Step 13: Open your application

Open **Chrome** and enter:

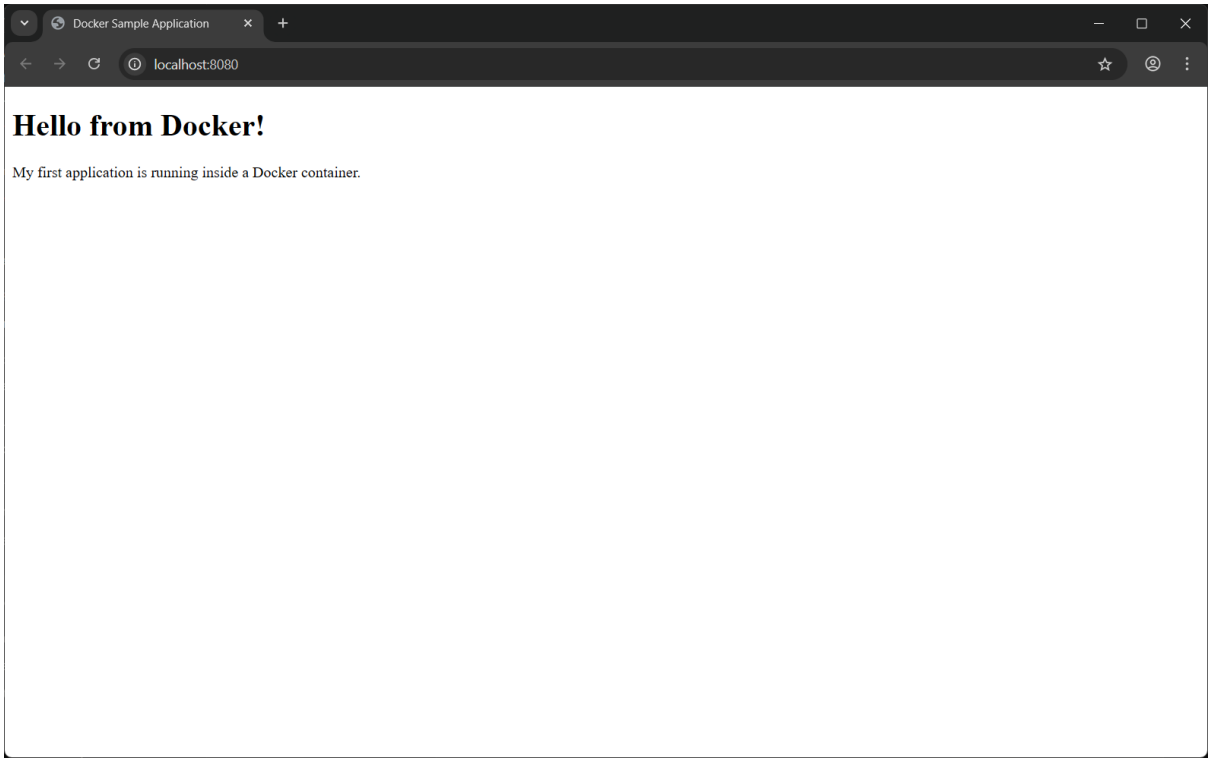
http://localhost:8080

You should see:

Hello from Docker!

and:

My first application is running inside a Docker container.



Step 14: Check the container

Go back to CMD and run:

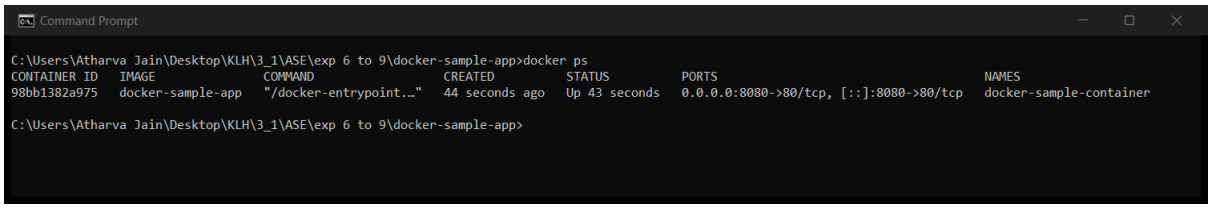
docker ps

You should see your container:

docker-sample-container

with status:

Up ...

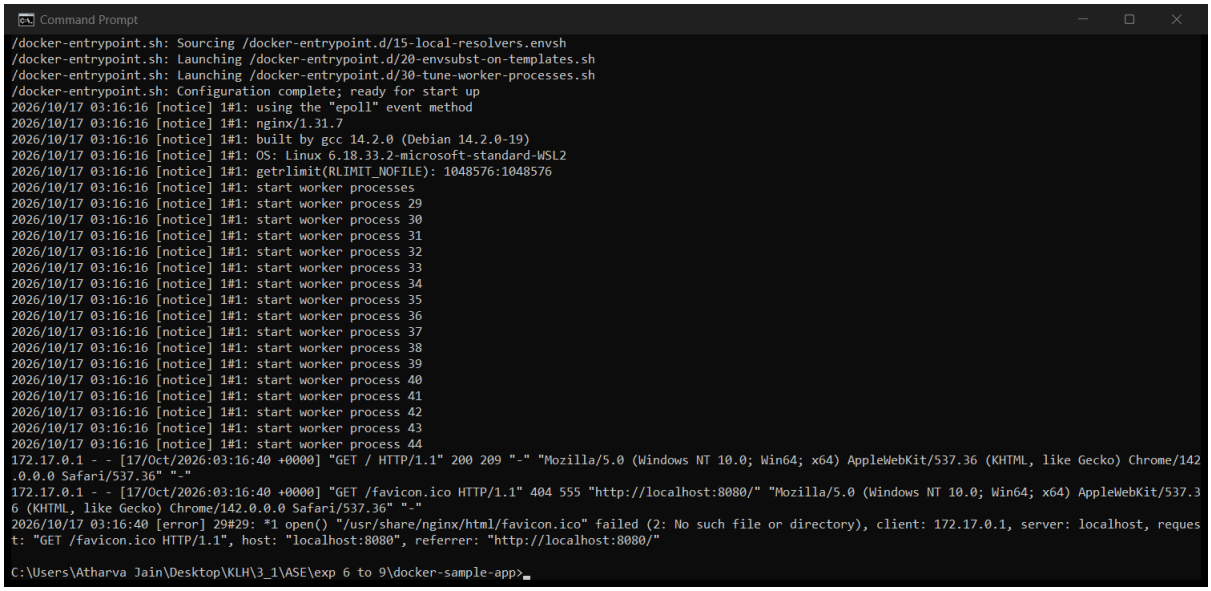


Step 15: Check the container logs

Run:

docker logs docker-sample-container

This displays the NGINX container logs.



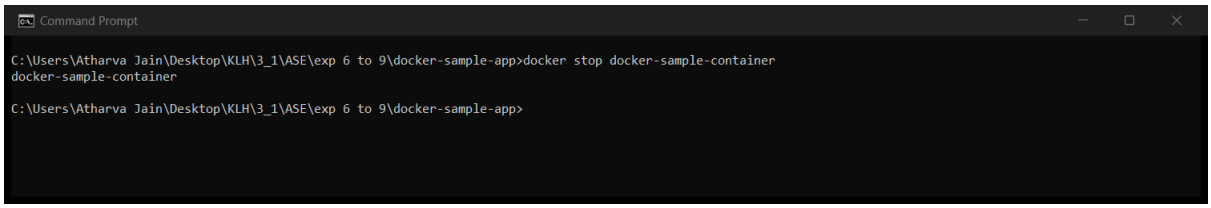
Step 16: Stop the container

After you have verified the webpage, run:

docker stop docker-sample-container

You should get:

docker-sample-container



Step 17: Check stopped containers

Run:

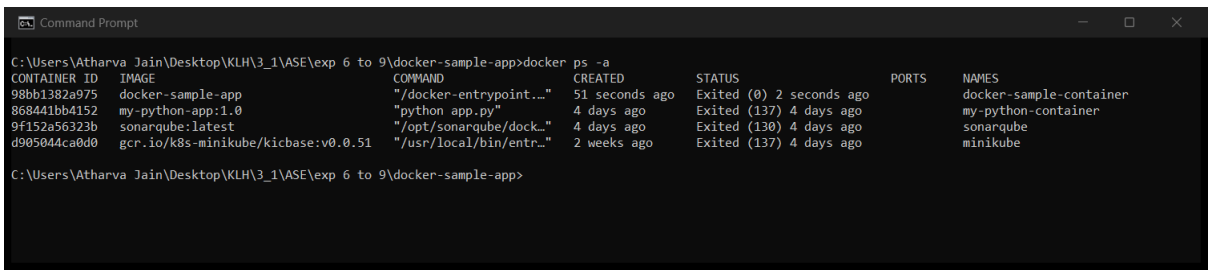
docker ps -a

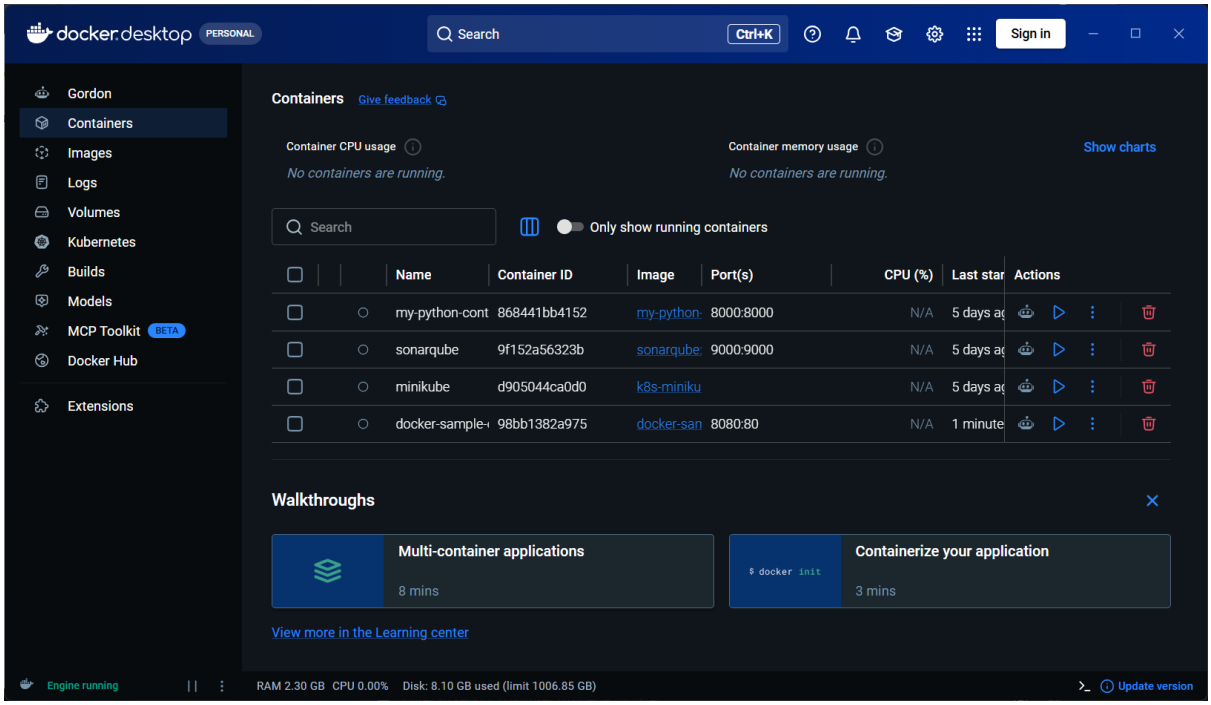
You should see:

docker-sample-container

with a status similar to:

Exited




docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Containers
Give feedback
Container CPU usage
No containers are running.
Container memory usage
No containers are running.
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last star
Actions
my-python-cont
868441bb4152
my-python-
8000:8000
N/A
5 days ag
sonarqube
9f152a56323b
sonarqube:
9000:9000
N/A
5 days ag
minikube
d905044ca0d0
k8s-miniku
N/A
5 days ag
docker-sample-
98bb1382a975
docker-san
8080:80
N/A
1 minute
Walkthroughs
Multi-container applications
8 mins
$ docker init
Containerize your application
3 mins
View more in the Learning center
Engine running
RAM 2.30 GB CPU 0.00% Disk: 8.10 GB used (limit 1006.85 GB)
Update version